

/* Query 1 - query used for first insight */

```
SELECT
  f.title AS film_title
,c.name AS category_name
,(SELECT
  COUNT(*)
  FROM rental r
  JOIN inventory i2
    ON r.inventory_id = i2.inventory_id
  WHERE i2.film_id = f.film_id)
  AS times_rented
FROM film f
JOIN film_category fc
  ON f.film_id = fc.film_id
JOIN category c
  ON fc.category_id = c.category_id
WHERE c.name IN (
'Animation', 'Children', 'Classics',
'Comedy', 'Family', 'Music'
)
ORDER BY times_rented DESC, film_title;
```

/* Query 2 - query used for second insight */

```
WITH rental_durations
AS
(SELECT
  f.film_id
,f.title
,c.name AS category_name
,r.rental_id
,DATE_PART('day', r.return_date - r.rental_date) AS rental_days
```

```
FROM film f
JOIN film_category fc
  ON f.film_id = fc.film_id
JOIN category c
  ON fc.category_id = c.category_id
JOIN inventory i
  ON f.film_id = i.film_id
JOIN rental r
  ON i.inventory_id = r.inventory_id
WHERE r.return_date IS NOT NULL),
```

avg_durations

AS

```
(SELECT
  film_id
, title
, category_name
, AVG(rental_days) AS avg_rental_days
FROM rental_durations
GROUP BY film_id
       , title
       , category_name),
```

film_with_quartile

AS

```
(SELECT
  film_id
, title
, category_name
, avg_rental_days
, NTILE(4) OVER (ORDER BY avg_rental_days) AS quartile_num
FROM avg_durations),
```

family_movies

AS

```
(SELECT
  film_id
, title
, category_name
, avg_rental_days
, CASE quartile_num
```

```

        WHEN 1 THEN 'first_quarter'
        WHEN 2 THEN 'second_quarter'
        WHEN 3 THEN 'third_quarter'
        WHEN 4 THEN 'final_quarter'
    END AS rental_quartile
FROM film_with_quartile
WHERE category_name IN (
    'Animation', 'Children', 'Classics',
    'Comedy', 'Family', 'Music'
))
SELECT
    film_id
    ,title
    ,category_name
    ,avg_rental_days
    ,rental_quartile
FROM family_movies
ORDER BY rental_quartile,
avg_rental_days,
title;

```

/* Query 3 - query used for third insight */

```

WITH film_avg
AS
(SELECT
    f.film_id
    ,f.title
    ,c.name AS category_name
    ,AVG(DATE_PART('day', r.return_date - r.rental_date)) AS
avg_rental_days
FROM film f
JOIN film_category fc
    ON f.film_id = fc.film_id

```

```
JOIN category c
  ON fc.category_id = c.category_id
JOIN inventory i
  ON f.film_id = i.film_id
JOIN rental r
  ON i.inventory_id = r.inventory_id
WHERE r.return_date IS NOT NULL
GROUP BY f.film_id
        ,f.title
        ,c.name),
```

```
film_with_quartile
AS
(SELECT
  film_id
  ,title
  ,category_name
  ,avg_rental_days
  ,NTILE(4) OVER (ORDER BY avg_rental_days) AS quartile_num
FROM film_avg),
```

```
film_with_bucket
AS
(SELECT
  film_id
  ,title
  ,category_name
  ,avg_rental_days
  ,CASE quartile_num
    WHEN 1 THEN 'first_quarter'
    WHEN 2 THEN 'second_quarter'
    WHEN 3 THEN 'third_quarter'
    WHEN 4 THEN 'final_quarter'
  END AS rental_length_category
FROM film_with_quartile),
```

```
family_only
AS
```

```
(SELECT
```

```
*
```

```
FROM film_with_bucket
```

```
WHERE category_name IN (
```

```
'Animation', 'Children', 'Classics',
```

```
'Comedy', 'Family', 'Music'
```

```
)),
```

```
family_with_counts
```

```
AS
```

```
(SELECT
```

```
category_name AS category
```

```
,rental_length_category
```

```
,COUNT(*) OVER (
```

```
PARTITION BY category_name, rental_length_category
```

```
) AS movie_count
```

```
FROM family_only)
```

```
SELECT DISTINCT
```

```
category
```

```
,rental_length_category AS "rental length category"
```

```
,movie_count AS "count"
```

```
FROM family_with_counts
```

```
ORDER BY category,
```

```
rental_length_category;
```

```
/* Query 4 - query used for forth insight */
```

```
SELECT
```

```
MONTH(r.rental_date) AS rental_month
```

```
,YEAR(r.rental_date) AS rental_year
```

```
,s.store_id
```

```
,COUNT(r.rental_id) AS count_rentals
```

```
FROM rental r
```

```
JOIN staff st
  ON r.staff_id = st.staff_id
JOIN store s
  ON st.store_id = s.store_id
GROUP BY YEAR(r.rental_date)
        ,MONTH(r.rental_date)
        ,s.store_id
ORDER BY count_rentals DESC;
```